CHAPTER 05

S3 & CloudFront

Object Storage · CDN · Security · Lifecycle · Lambda@Edge

05-00 Chapter Overview

Topic	AWS Service	ShopFlow Use-Case
S3 Storage Classes	Standard, IA, Glacier, Intelligent-Tiering	Product images Standard → IA after 90 days
S3 Static Website	S3 + CloudFront + Route53	React SPA hosted in S3 behind CloudFront
S3 Security	Bucket policies, ACLs, S3 Block Public Access	All buckets private — CloudFront OAC only
S3 Lifecycle Rules	Transition + Expiration rules	Auto-tier product images, delete tmp files 7d
S3 Event Notifications	SQS/SNS/Lambda triggers on object events	Trigger thumbnail generation on upload
S3 Replication	CRR (cross-region) + SRR (same-region)	CRR to us-west-2 for disaster recovery
S3 Performance	Multipart, S3 Transfer Acceleration, prefix sharding	Multipart for images > 5MB
CloudFront CDN	Edge locations, Origins, Behaviors, Cache Policies	600+ PoPs serve ShopFlow global customers
CloudFront Security	WAF integration, Geo Restriction, Signed URLs	Signed URLs for purchase receipts, WAF on API
CloudFront Functions	Viewer Request/Response functions (JS)	A/B test, URL normalization at edge
Lambda@Edge	Viewer/Origin Request/Response (Node.js)	Auth header injection, dynamic resizing
S3 Cost Optimization	Intelligent-Tiering, Glacier Deep Archive	Save 73% on archived order attachments

05-01 S3 Core Concepts — Buckets, Objects & Consistency

Amazon S3 is ShopFlow's primary object store for product images (12M objects, 4.2TB), static React SPA assets (220MB), order PDF receipts (1.8M docs/year), CloudWatch logs archive, and Aurora automated backups. S3 provides 11 nines (99.999999999%) durability — data is replicated across ≥3 AZs within a region.

S3 Core Concepts

Concept	Description	ShopFlow Example
Bucket	Globally unique container for objects. Region-scoped.	shopflow-product-images-prod (us-east-1)
Object	File + metadata. Max 5TB. Key is full path.	products/electronics/iphone15/main.jpg
Object Key	Unique identifier within a bucket (the full path).	receipts/2024/11/29/order-8827361.pdf
Versioning	Keep all versions of an object. Protects against accidental deletes.	Enabled on product-images bucket (rollback)
Consistency Model	Strong read-after-write for all operations (since 2020).	After PUT completes, GET immediately returns new object
Storage Class	Tier that determines durability, availability, and price.	Standard for active images, IA after 90 days
Prefix	String before the first "/" — NOT a real directory.	products/electronics/ — shard by prefix for performance
ETag	MD5 hash of object content — used for conditional requests and cache validation.	CloudFront uses ETag for cache invalidation

ShopFlow S3 Bucket Inventory

Bucket Name	Contents	Size	Storage Class	Versioning
shopflow-product-images-prod	All product photos, thumbnails	4.2 TB	Standard → IA (90d)	Yes
shopflow-spa-assets-prod	React build: JS, CSS, HTML, fonts	220 MB	Standard	No
shopflow-order-receipts-prod	PDF receipts per order	890 GB (growing)	Standard → Glacier (1yr)	Yes
shopflow-logs-archive-prod	ALB + CloudFront access logs	2.1 TB / year	Standard → Glacier IA (30d)	No
shopflow-db-backups-prod	Aurora automated snapshots export	1.8 TB	Standard → Deep Archive (90d)	Yes
shopflow-cf-templates-prod	CDK / CloudFormation templates	15 MB	Standard	Yes

■ Use CloudFront Origin Access Control (OAC) — not presigned URLs and not public buckets — for serving S3 content. OAC requires no client-side signing, is IAM-controlled, and supports all S3 request types including SSE-KMS objects.

05-02 S3 Storage Classes — Cost vs Access Tradeoff

S3 offers 6 storage class tiers with different durability, availability, access latency, and price. Choosing the right class for each object's access pattern is the single biggest S3 cost lever. ShopFlow saves ~$580/month by tiering rarely-accessed content.

Storage Class	Durability	Availability	Min Duration	$/GB/mo	Retrieval Fee	ShopFlow Usage
S3 Standard	11 nines	99.99% (3 AZ)	None	$0.023	None	Active product images, SPA ass
S3 Intelligent-Tiering	11 nines	99.9%	None	$0.023 + $0.0025/1K obj/mo	None	Product images with unknown ac
S3 Standard-IA	11 nines	99.9% (3 AZ)	30 days	$0.0125	$0.01/GB	Product images 90-365 days old
S3 One Zone-IA	11 nines	99.5% (1 AZ only)	30 days	$0.01	$0.01/GB	Reproducible data (thumbnails c
S3 Glacier Instant	11 nines	99.9%	90 days	$0.004	$0.03/GB	Order receipts older than 1 year
S3 Glacier Flexible	11 nines	99.99%	90 days	$0.0036	$0.01-$0.03/GB, 1-12hr	Order attachments 2+ years old
S3 Glacier Deep Archive	11 nines	99.99%	180 days	$0.00099	$0.02/GB, 12-48hr	DB backups beyond 90 days —

Lifecycle Policy — CDK

```
const imagesBucket = new s3.Bucket(this,"ProductImages",{
bucketName: "shopflow-product-images-prod",
versioned: true,
encryption: s3.BucketEncryption.S3_MANAGED,
blockPublicAccess: s3.BlockPublicAccess.BLOCK_ALL,
lifecycleRules: [{
id: "ImageTiering",
enabled: true,
transitions: [
{storageClass: s3.StorageClass.INFREQUENT_ACCESS,
transitionAfter: Duration.days(90)},
{storageClass: s3.StorageClass.GLACIER_INSTANT_RETRIEVAL,
transitionAfter: Duration.days(365)},
],
noncurrentVersionsToRetain: 3,
noncurrentVersionTransitions: [{
storageClass: s3.StorageClass.INFREQUENT_ACCESS,
transitionAfter: Duration.days(30),
}],
},{
id: "DeleteIncompleteMultipart",
abortIncompleteMultipartUploadAfter: Duration.days(7),
}],
});
```

■ Enable S3 Intelligent-Tiering for product images where access patterns are uncertain. It monitors 30-day access frequency automatically and moves objects between Frequent Access (Standard price) and Infrequent Access (IA price) tiers with no retrieval fees and no minimum duration penalty.

05-03 S3 Security — Zero Public Access Architecture

ShopFlow enforces a "zero public S3" policy. Every bucket has Block Public Access enabled at both account and bucket level. All content delivery goes through CloudFront with Origin Access Control (OAC), which cryptographically signs every S3 request using the CloudFront distribution's IAM-equivalent identity.

OAC Bucket Policy (CDK-generated)

```
imagesBucket.addToResourcePolicy(new iam.PolicyStatement({
effect: iam.Effect.ALLOW,
principals: [new iam.ServicePrincipal("cloudfront.amazonaws.com")],
actions: ["s3:GetObject"],
resources: [imagesBucket.arnForObjects("*")],
conditions: {
"StringEquals": {
"AWS:SourceArn": cfDistribution.distributionArn
}
}
}));
```

S3 Presigned URL — Lambda Handler

```
// Generate presigned URL for order receipt download
const command = new GetObjectCommand({
Bucket: "shopflow-order-receipts-prod",
Key: `receipts/${year}/${month}/${orderId}.pdf`,
});
const presignedUrl = await getSignedUrl(s3Client, command, {
expiresIn: 900, // 15 minutes — order receipt is a one-time download
```

```
});
return { statusCode: 200, body: JSON.stringify({ downloadUrl: presignedUrl }) };
```

■ Never log S3 presigned URLs — they are bearer tokens. Anyone with the URL can access the object until expiry. Always set short expiry (5-15 minutes), log only the object key, and use bucket policy Conditions to restrict to specific source IPs if possible.

05-04 S3 Event Notifications — Thumbnail Generation Pipeline

S3 Event Notifications trigger SQS, SNS, or Lambda functions when objects are created, deleted, or restored. ShopFlow uses S3 → SQS → Lambda to generate product image thumbnails in 4 sizes when sellers upload new images.

S3 Event Notification — CDK

```
import * as s3n from "aws-cdk-lib/aws-s3-notifications";
imagesBucket.addEventNotification(
s3.EventType.OBJECT_CREATED,
new s3n.SqsDestination(thumbnailQueue),
{ prefix: "products/originals/", suffix: ".jpg" },
);
imagesBucket.addEventNotification(
s3.EventType.OBJECT_CREATED,
new s3n.SqsDestination(thumbnailQueue),
{ prefix: "products/originals/", suffix: ".png" },
);
// Lambda triggered by SQS (batch window: 30 seconds, batch size: 10)
thumbnailFn.addEventSource(new lambdaEvents.SqsEventSource(thumbnailQueue, {
batchSize: 10,
```

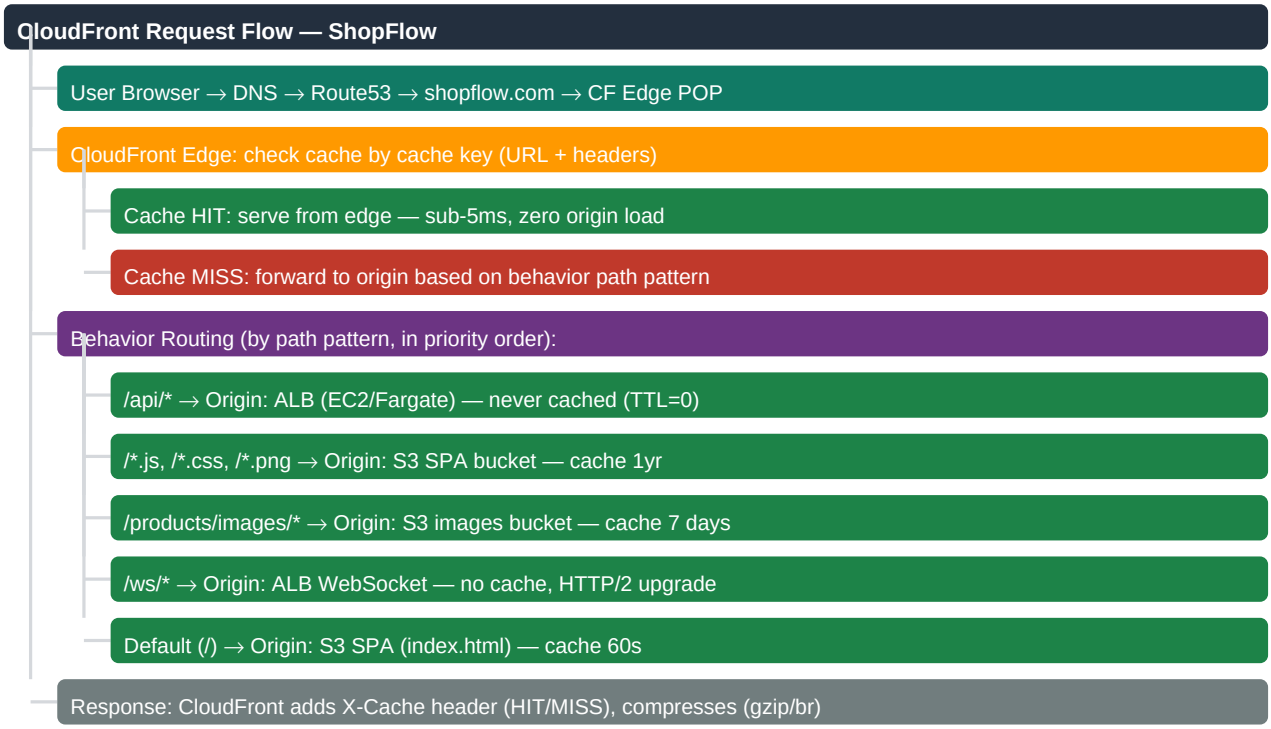
```
maxBatchingWindow: Duration.seconds(30),
reportBatchItemFailures: true, // partial batch failure support
}));
```

■ Use SQS as buffer between S3 events and Lambda processing. Direct S3 → Lambda can overwhelm Lambda concurrency during bulk uploads (sellers sometimes upload 500 images at once). SQS smooths the burst, and Lambda scales at a safe rate.

05-05 CloudFront CDN — Global Content Distribution

Amazon CloudFront is a global CDN with 600+ edge locations. ShopFlow uses CloudFront as the single entry point for all web traffic — the SPA, product images, APIs, and WebSocket connections — enabling global sub-50ms latency for static content and reducing ALB traffic by 85%.

CloudFront Request Lifecycle



CloudFront Distribution — CDK

```
const distribution = new cloudfront.Distribution(this,"ShopFlowCF",{
comment: "shopflow.com CDN — production",
defaultRootObject: "index.html",
domainNames: ["shopflow.com","www.shopflow.com"],
certificate: acmCert,
minimumProtocolVersion: cloudfront.SecurityPolicyProtocol.TLS_V1_2_2021,
enableLogging: true,
logBucket: logsBucket,
logFilePrefix: "cf-access-logs/",
// Default behavior: React SPA from S3
defaultBehavior: {
origin: new origins.S3BucketOrigin(spaBucket, {
originAccessControl: oac }), // OAC (not OAI — deprecated)
viewerProtocolPolicy: cloudfront.ViewerProtocolPolicy.REDIRECT_TO_HTTPS,
cachePolicy: cloudfront.CachePolicy.CACHING_OPTIMIZED,
compress: true,
},
additionalBehaviors: {
"/api/*": {
origin: new origins.LoadBalancerV2Origin(alb, {
protocolPolicy: cloudfront.OriginProtocolPolicy.HTTPS_ONLY }),
cachePolicy: cloudfront.CachePolicy.CACHING_DISABLED,
allowedMethods: cloudfront.AllowedMethods.ALLOW_ALL, // pass POST/PUT/DELETE
viewerProtocolPolicy: cloudfront.ViewerProtocolPolicy.HTTPS_ONLY,
},
```

```
},
});
```

■ Use CachePolicy.CACHING_DISABLED for /api/* (TTL=0, no caching) but still benefit from CloudFront: global anycast routing routes requests to the nearest AWS edge, then AWS backbone to your ALB — faster than public internet even for dynamic content.

05-06 CloudFront Security — WAF, Signed URLs & Geo Restriction

CloudFront integrates with AWS WAF for application-layer protection, supports Signed URLs and Signed Cookies for private content, and offers Geo Restriction to block traffic from specific countries.

CloudFront Security Layers

Security Layer	What It Blocks	ShopFlow Config
AWS WAF on CloudFront	SQL injection, XSS, rate limiting, bot traffic	AWSManagedRulesCommonRuleSet + custom rate rules
HTTPS Only (minimum TLSv1.2)	HTTP downgrade attacks, BEAST, POODLE	TLS_V1_2_2021 minimum policy — drops TLS 1.0/1.1
CloudFront Signed URLs	Unauthorized access to private S3 content	Order receipts: CF signed URL, 15-min TTL
Geo Restriction (block list)	Traffic from specific countries (export control)	Block embargoed countries — OFAC compliance
Origin Shield	Reduces origin load, adds caching layer	Enabled for S3 origin in us-east-1
Field-Level Encryption	Protects sensitive form fields end-to-end	Payment form: RSA encrypt card number at edge

WAF WebACL — CDK (CloudFront scope)

```
// WAF rules for CloudFront MUST be created in us-east-1 (global scope)
const webAcl = new wafv2.CfnWebACL(this,"ShopFlowWAF",{
scope: "CLOUDFRONT", // must be CLOUDFRONT or REGIONAL
defaultAction: {allow: {}},
rules: [{
name: "AWSCommonRules",
priority: 1,
overrideAction: {none: {}},
statement: {managedRuleGroupStatement: {
vendorName: "AWS",
name: "AWSManagedRulesCommonRuleSet"}},
visibilityConfig: {sampledRequestsEnabled:true,
cloudWatchMetricsEnabled:true,
metricName:"CommonRules"},
},{
name: "CheckoutRateLimit",
priority: 2,
action: {block: {}},
statement: {rateBasedStatement: {
limit: 100, aggregateKeyType:"IP",
scopeDownStatement: {byteMatchStatement:{
fieldToMatch:{uriPath:{}},
positionalConstraint:"STARTS_WITH",
searchString:"/api/v1/orders",
textTransformations:[{priority:0,type:"NONE"}]}}},
visibilityConfig:{sampledRequestsEnabled:true,
cloudWatchMetricsEnabled:true,metricName:"CheckoutRateLimit"},
}],
visibilityConfig:{sampledRequestsEnabled:true,
cloudWatchMetricsEnabled:true,metricName:"ShopFlowWAF"},
});
```

■ WAF for CloudFront MUST be created in us-east-1 region — CloudFront is a global service tied to us-east-1. Even if your application is in us-west-2, create the WebACL stack in us-east-1 with env: { region: "us-east-1" } in CDK.

05-07 Lambda@Edge — Dynamic Processing at Edge Locations

Lambda@Edge runs Node.js or Python functions at CloudFront edge locations — reducing latency for dynamic processing that'd otherwise require a round-trip to the origin. ShopFlow uses Lambda@Edge for auth header injection and dynamic image resizing.

Lambda@Edge Trigger Points

Trigger	Timing	Latency Impact	Max Execution Time	ShopFlow Use
Viewer Request	After CF receives request, before cache check	Highest impact — runs every request (cache hit+miss)	5 seconds	URL normalization: /products → /pr
Origin Request	Before request goes to origin (cache miss only)	Medium — cache misses only	30 seconds	Inject auth header for ALB origin va
Origin Response	After origin responds, before CF caches	Medium — cache misses only	30 seconds	Add security headers (HSTS, CSP)
Viewer Response	After CF returns to viewer (every request)	Highest — runs every response	5 seconds	Not used (use Origin Response ins

Lambda@Edge — Auth Header Injection (Origin Request)

```
// TypeScript Lambda@Edge — Origin Request trigger
// Injects X-CloudFront-Secret header to verify request came via CF, not direct
export const handler = async (event: CloudFrontRequestEvent) => {
const request = event.Records[0].cf.request;
// Inject shared secret that ALB security group only trusts from CF
request.headers['x-cloudfront-secret'] = [{
key: 'X-CloudFront-Secret',
value: process.env.CF_SECRET! // from Lambda@Edge env (us-east-1 only)
}];
// For image paths, rewrite to optimal format based on Accept header
if (request.uri.startsWith('/products/images/')) {
const acceptHeader = request.headers['accept']?.[0]?.value ?? '';
if (acceptHeader.includes('image/webp')) {
request.uri = request.uri.replace(/\.jpg$/, '.webp'); // serve WebP
}
}
return request;
};
```

CloudFront Functions vs Lambda@Edge — Quick Choice Guide

Feature	CloudFront Functions	Lambda@Edge
Languages	JavaScript (ES5)	Node.js, Python
Max execution time	1ms (!!)	5-30 seconds
Access to network/filesystem	No — pure computation only	Yes (DynamoDB, Secrets Manager)
Price vs Lambda@Edge	~10x cheaper	Standard Lambda pricing
Trigger points	Viewer Request/Response only	All 4 trigger points
Use for	URL rewrite, header manipulation, A/B routing	Auth, DB lookup, image transform

■ Use CloudFront Functions for simple request manipulation (URL normalization, header add/remove, A/B testing by cookie). Use Lambda@Edge only when you need network access or >1ms execution time. CloudFront Functions cost $0.10/million vs Lambda@Edge at $0.60/million.

05-08 S3 Performance — Multipart Upload & Transfer Acceleration

S3 throughput scales with prefix diversity and upload strategy. For ShopFlow's large image uploads (some RAW product photos are 50MB+), multipart upload is required for files over 5GB and recommended for files over 5MB.

Multipart Upload — When and How

File Size	Recommended Strategy	Benefit
< 5 MB	Single PUT upload	Simpler, lower overhead for small files
5 MB - 100 MB	Multipart upload (10 MB parts)	Retry individual parts, parallel upload
100 MB - 1 GB	Multipart upload (25 MB parts)	4 parallel streams = 4x throughput
> 1 GB	Multipart + S3 Transfer Acceleration	TA routes via AWS edge → backbone, much faster from far regions
> 5 GB (required)	Multipart is mandatory — single PUT max is 5 GB	S3 API requirement, not optional

Multipart Upload — Spring Boot

```
// Upload product image with multipart (Spring Boot + AWS SDK v2)
@Service
public class ProductImageService {
private final S3Client s3;
private static final int PART_SIZE = 10 * 1024 * 1024; // 10 MB parts
public String uploadProductImage(String productId, InputStream imageStream,
long fileSize, String contentType) {
String key = "products/originals/" + productId + "/main.jpg";
CreateMultipartUploadResponse init = s3.createMultipartUpload(
r -> r.bucket("shopflow-product-images-prod").key(key)
.contentType(contentType)
.serverSideEncryption(ServerSideEncryption.AES256));
List<CompletedPart> parts = new ArrayList<>();
byte[] buffer = new byte[PART_SIZE];
int partNumber = 1; int bytesRead;
while ((bytesRead = imageStream.read(buffer)) > 0) {
UploadPartResponse part = s3.uploadPart(
r -> r.bucket("shopflow-product-images-prod").key(key)
.uploadId(init.uploadId()).partNumber(partNumber),
RequestBody.fromBytes(Arrays.copyOf(buffer, bytesRead)));
parts.add(CompletedPart.builder().partNumber(partNumber++)
.eTag(part.eTag()).build());
}
s3.completeMultipartUpload(r -> r
.bucket("shopflow-product-images-prod").key(key)
.uploadId(init.uploadId())
.multipartUpload(m -> m.parts(parts)));
return "https://cdn.shopflow.com/" + key;
}
}
```

■ Add an S3 lifecycle rule to abort incomplete multipart uploads after 7 days: abortIncompleteMultipartUploadAfter: Duration.days(7). Failed uploads that don't get completed leave orphaned parts that incur storage charges indefinitely.

05-09 S3 Replication — Cross-Region DR

S3 Cross-Region Replication (CRR) asynchronously copies every new object from a source bucket to a destination bucket in a different region. ShopFlow replicates product images to us-west-2 for disaster recovery — RTO <15 minutes for image serving failover.

CRR vs SRR — When to Use Each

Replication Type	Source → Destination	Latency	Use Case	Cost
Cross-Region (CRR)	Different AWS regions	Minutes (async, best-effort)	DR, compliance, multi-region serving	Storage in both regions + transfe
Same-Region (SRR)	Same region, different bucket	Seconds (async)	Log aggregation, account separation, test/prod copy	Storage in both buckets, no trans
S3 Batch Replication	Any → any (existing objects)	Hours to days	Backfill existing objects that predate replication	Per-object fee ($0.25/million obje
Bi-directional CRR	Region A ↔ Region B (both directions)	Minutes each way	Active-active multi-region (requires conflict handling)	Double transfer + double storage

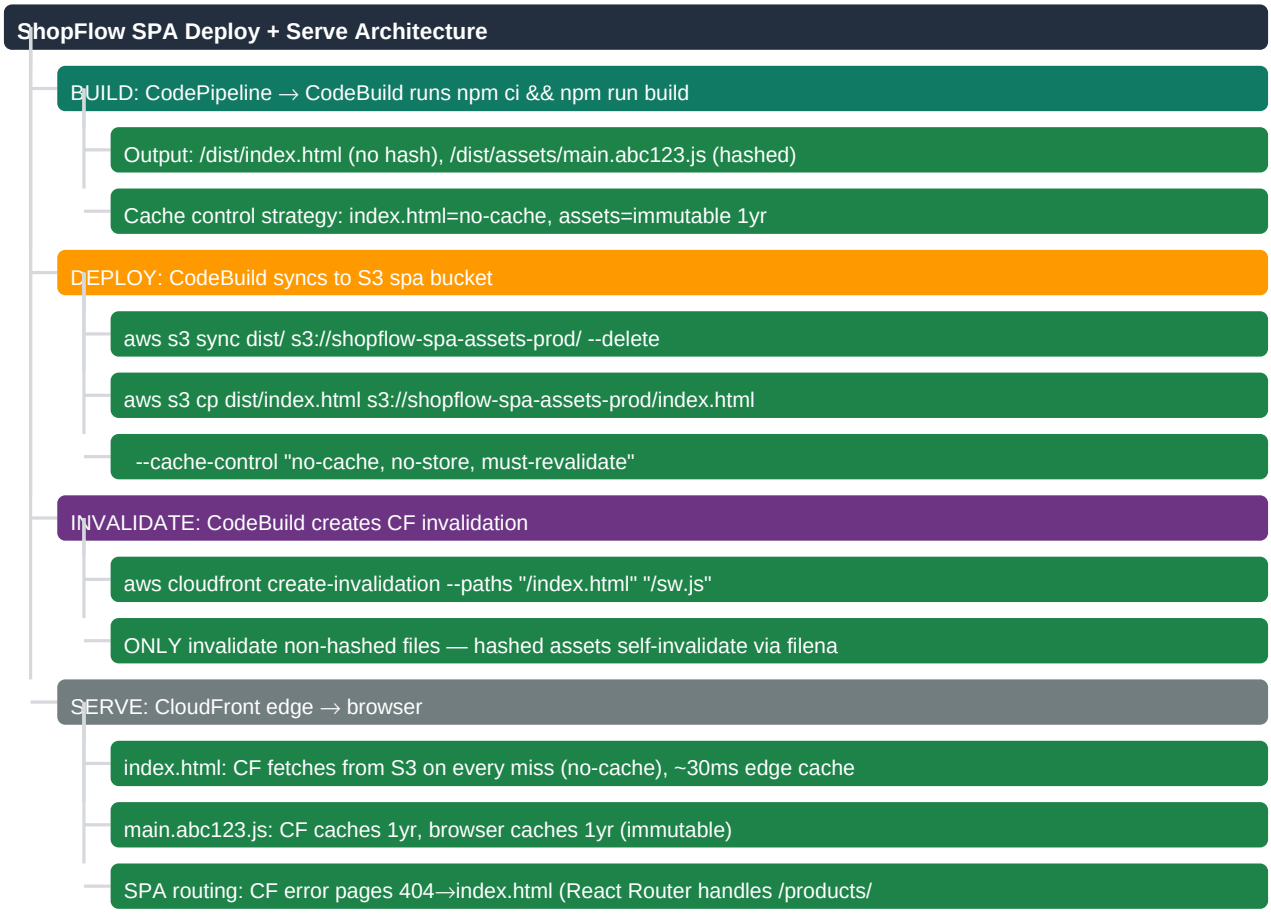
CRR CDK Configuration

```
const sourceRegion = "us-east-1";
const destRegion = "us-west-2";
// Source bucket (us-east-1) — versioning REQUIRED for replication
const sourceBucket = new s3.Bucket(this,"SourceImages",{
bucketName: "shopflow-product-images-prod",
versioned: true, // REQUIRED for CRR
replicationRules: [{
id: "ReplicateToDR",
status: s3.ReplicationRuleStatus.ENABLED,
destination: {
bucket: drBucket, // in us-west-2
storageClass: s3.StorageClass.STANDARD_IA, // save cost in DR
},
filter: {prefix: "products/"}, // only replicate product images
deleteMarkerReplication: false, // do NOT replicate deletes
}],
});
// Enable S3 Replication Time Control (RTC) for 99.99% replication in 15min
// (required for SLA compliance — additional $0.015/GB replicated)
```

■ Disable deleteMarkerReplication unless you specifically need deletes replicated. With it enabled, a developer accidentally running "delete all" in production would propagate to your DR bucket. Keep delete protection in DR bucket as an independent safety layer.

05-10 ShopFlow Static SPA Hosting Architecture

ShopFlow's React frontend is a Single Page Application (SPA) deployed to S3 and served via CloudFront. The deployment pipeline runs on CodePipeline, invalidates CloudFront cache on each deploy, and uses immutable hashed filenames for JS/CSS to enable aggressive 1-year caching.



CloudFront Error Page for SPA Routing

```
// SPA routing: 404 from S3 → return index.html (React Router takes over)
const distribution = new cloudfront.Distribution(this,"ShopFlowCF",{
errorResponses: [{
httpStatus: 403, // S3 returns 403 for missing objects (not 404)
responseHttpStatus: 200,
responsePagePath: "/index.html",
ttl: Duration.seconds(0), // do not cache error responses
},{
httpStatus: 404,
responseHttpStatus: 200,
responsePagePath: "/index.html",
ttl: Duration.seconds(0),
}],
// ... rest of distribution config
});
```

■ S3 returns 403 (not 404) for missing objects when Block Public Access is enabled — there's no "file not found" since S3 won't reveal whether the key exists. Map both 403 AND 404 to /index.html in CloudFront error responses for reliable SPA routing.

05-11 S3 Intelligent-Tiering & Cost Analysis

S3 storage costs account for ~$340/month of ShopFlow's AWS bill. Smart tiering reduces this by 68% without requiring manual lifecycle rule tuning for objects with unknown or changing access patterns.

ShopFlow S3 Monthly Cost — Before vs After

Bucket / Content Type	Volume	Before Optimization	After Tiering	Monthly Saving
Product images (Standard)	4.2 TB	$96.60/mo	$60 (IA after 90d, Glacier 1yr)	$36.60
Order receipts	890 GB	$20.47/mo	$6.75 (IA 1yr, Glacier 2yr)	$13.72
Log archive (Standard)	2.1 TB	$48.30/mo	$9.22 (Deep Archive after 30d)	$39.08
DB backups (Standard)	1.8 TB	$41.40/mo	$1.78 (Deep Archive after 90d)	$39.62
SPA assets (Standard)	220 MB	$0.005/mo	$0.005 (already tiny)	$0
Requests + data transfer	Varies	$133/mo	$133 (unchanged)	$0
Total	~9 TB	$340/month	$211/month	$129/month (38% saved)

S3 Intelligent-Tiering — How It Works

IT Sub-Tier	Activation Condition	Monthly Cost	Retrieval Fee
Frequent Access	Default for new objects	$0.023/GB (Standard price)	None
Infrequent Access	No access for 30 consecutive days	$0.0125/GB (IA price)	None
Archive Instant Access	No access for 90 consecutive days	$0.004/GB (Glacier Instant)	None
Archive Access (optional)	No access for 90-270 days (opt-in)	$0.0036/GB (Glacier Flexible)	$0.01/GB
Deep Archive Access (optional)	No access for 180+ days (opt-in)	$0.00099/GB (Deep Archive)	$0.02/GB

S3 Data Transfer Cost Optimization

- S3 to CloudFront: $0.00/GB (free — S3 → CloudFront is always free)
- S3 to internet (direct): $0.09/GB first 10TB/month — always use CloudFront instead
- S3 to EC2 same region: $0.00/GB (free within same region)
- S3 to EC2 different region: $0.09/GB — always use same region for compute and storage
- S3 Cross-Region Replication transfer: $0.02/GB (us-east-1 → us-west-2)

■ Enable S3 Storage Lens for organization-wide storage analytics. It provides 60+ metrics including access patterns per bucket, incomplete multipart uploads wasting space, and storage class recommendations — free tier covers 14 days of metrics.

05-BP Best Practices & Anti-Patterns

✓ Enforce S3 Block Public Access at the AWS account level — prevents any future bucket from going public accidentally

✓ Use CloudFront OAC (Origin Access Control) instead of OAI (deprecated) or public buckets

✓ Set abort-incomplete-multipart-upload lifecycle rule (7 days) on all buckets to prevent orphaned parts

✓ Use hashed filenames for static assets (main.abc123.js) — enables 1-year CloudFront cache with no invalidation cost

✓ Enable S3 Versioning on critical buckets — protects against accidental deletion and ransomware overwrites

✓ Use S3 presigned URLs (15min TTL) for private content downloads — not permanent public URLs

✓ Enable S3 server-side encryption (SSE-S3 minimum, SSE-KMS for compliance) on all buckets

✓ Use CloudFront with Origin Shield to reduce origin cache-miss requests by additional 60-80%

■ Never use S3 presigned URLs for CDN content — CloudFront OAC is faster, cheaper, and more secure

■ Do not use ACLs — use bucket policies only. AWS recommends disabling ACLs completely on all new buckets

■ Do not serve S3 directly (bypass CloudFront) — no edge caching, no WAF, higher data transfer cost ($0.09/GB vs $0.00/GB)

■ S3 WAF is only available on CloudFront (not S3 directly) — always put WAF on CloudFront, not the origin

■ Do not over-invalidate CloudFront — each path invalidation after first 1,000/month costs $0.005. Use versioned filenames instead

05-QR Quick Reference & Interview Q&A

Concept	Key Fact
S3 object max size	5 TB per object (5 GB max for single PUT — multipart required above 5GB)
S3 durability	11 nines (99.999999999%) — data replicated across ≥3 AZs
S3 consistency	Strong read-after-write for all operations (since December 2020)
CloudFront edge count	600+ edge locations in 100+ cities, 50+ countries
CloudFront to S3 transfer cost	$0.00/GB — free between AWS services in same account
S3 → internet direct cost	$0.09/GB — always use CloudFront instead
WAF for CloudFront region	Must create WebACL in us-east-1 (CloudFront is global, tied to us-east-1)
Multipart required at	5 GB (single PUT max). Recommended at 5 MB for retry + parallel benefits
Intelligent-Tiering monitoring fee	$0.0025/1,000 objects/month — not worth it for objects < 128KB
S3 Glacier Deep Archive price	$0.00099/GB/month — cheapest durable AWS storage (~$1/TB/month)
OAC vs OAI	OAC (new): supports SSE-KMS, all S3 APIs. OAI (deprecated): no KMS, limited. Always use OAC
Lambda@Edge vs CF Functions	CF Functions: JS only, 1ms max, cheaper ($0.10/M). L@Edge: Node/Python, 30s, network access ($0.
S3 CRR requirement	Both source and destination buckets MUST have Versioning enabled

Top Interview Questions

Question	Answer
How does CloudFront OAC work?	OAC uses SigV4 to sign every S3 request with the CloudFront distribution's identity (cloudfront.amazon
S3 vs EBS vs EFS — when to use each?	S3: object storage, unlimited scale, REST API, no mounting. Best for files, images, backups, static web
How do you make CloudFront serve an SPA?	S3 bucket + CloudFront distribution. Set defaultRootObject=index.html. Map 403/404 error responses
What is S3 Intelligent-Tiering and how it works?	Automatically moves objects between Frequent Access (Standard price), Infrequent Access, and Glacie
How do you handle Black Friday traffic with S3/CloudFront?	CloudFront caches with 99.9%+ hit ratio at edge locations globally — no action needed for scale. Pre-warm
S3 CRR vs SRR?	CRR (Cross-Region Replication): source and destination in different regions. For DR and compliance. I
How to secure S3 from public access?	Enable S3 Block Public Access at account + bucket level. Use OAC for CloudFront. Use presigned URI